Calcurion Physics Worksheets

A-Level Physics

Topic: Electricity

MARKSCHEME

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1. Rate of flow of charge **A1** [1]
2. n number density, A area, v drift speed, q charge **A3** [3]
3. $v = I/(nAe) = 1.47 \times 10^{-4}$ m/s **M1A1** [3]
4. Work done per unit charge **A1** [1]
5. $V \propto I$ at constant temperature **A2** [2]
6. $R = 24\ \Omega$; $P = 1.5$ W **M1A1** [3]
7. $\rho = RA/L$; Ω m **A2** [2]
8. $\rho = 2.0 \times 10^{-7}\ \Omega$ m **M1A1** [3]
9. Sum of currents into a junction = sum out **A1** [1]
10. Sum of e.m.f.s = sum of p.d.s in a closed loop **A2** [2]
11. $I = 1.2$ A; $V_4 = 4.8$ V, $V_6 = 7.2$ V **M1A1** [3]
12. $R = 2.4\ \Omega$; $I = 5.0$ A **M1A1** [3]
13. Energy converted per unit charge from chemical to electrical **A1** [1]
14. $I = 0.50$ A; $V = 1.25$ V **M1A1** [3]
15. Conducts one way; threshold $\sim$0.6 V for Si **A2** [2]
16. $P = IV = I(IR) = I^2R$ **A2** [2]
17. $R = 0.34\ \Omega$; $P = 34$ W **M1A1** [4]
18. $\mathcal{E} = I(R + r) = IR + Ir = V + Ir$ **A3** [3]
19. $V_{\text{out}} = 6.0$ V **M1A1** [3]
20. Lattice vibrations increase; more collisions; v decreases **A3** [3]
21. Parallel = 1.2 Ω; total = 5.2 Ω; $I = 2.3$ A **M1A1** [4]
22. $Q = 18$ C **M1A1** [2]
23. Parallel = 2.5 Ω; total $R = 3.5\ \Omega$; $I = 3.43$ A **M1A1** [4]
24. 1 kWh = 3.6 MJ; cost = 1.13 **M1A1** [3]
25. Measure R, L, d; $\rho = RA/L$ **A4** [4]